

Technical Engineering Team · 2025

Enterprise Database Investigation & Architecture Reflection

Oracle · SQL Server · PostgreSQL · MySQL

1. DBMS Comparison
2. Oracle Architecture
3. Enterprise Decision Thinking
4. Architecture & Scalability Analysis
5. Migration & Industry Reflection

Part 1 — DBMS Technologies Investigation

DBMS	Type	Industries	Key Strengths	Trade-offs	Licensing
Oracle	Commercial	Banking Healthcare Telecom Gov	RAC multi-node HA Advanced security PL/SQL, Flashback	Very expensive Vendor lock-in Complex admin	\$47,500+/processor XE = free
SQL Server	Commercial	Finance Retail Manufacturing Education	Azure integration Power BI/SSRS/SSIS T-SQL, AlwaysOn AG	Windows-first High EE cost Microsoft lock-in	EE \$14K+/core Express = free
PostgreSQL	Open-source (BSD)	Startups, SaaS Fintech Analytics AI/ML	Free, ACID, JSONB Extensible (PostGIS, pgvector) Standard SQL	No built-in RAC HA needs extra tools No vendor SLA	Free (BSD) EDB for enterprise
MySQL	Open-source (GPL)	Web apps E-commerce CMS Social media	Easy setup Fast reads LAMP stack Huge community	Weaker SQL compliance Limited advanced features Oracle ownership concern	Community = free Enterprise \$5K–10K/yr

Enterprise vs Open-source: Oracle and SQL Server are commercial with vendor-backed SLAs — mandatory in regulated industries. PostgreSQL and MySQL are open-source but enterprise support requires third-party agreements (EDB / Percona).

Part 2 — Oracle Architecture & Versions

Oracle Editions Comparison

Feature	XE (Express)	Standard Edition 2	Enterprise Edition
Cost	Free	~\$17,500/socket	\$47,500+/processor
Target use	Dev / learning only	Small–medium business	Banks, telecoms, gov
Max CPU	1 CPU, 2 threads	2 sockets	Unlimited
Max RAM	2 GB	Unlimited	Unlimited
Max user data	12 GB	Unlimited	Unlimited
PDBs per CDB	1 PDB	1 PDB (Oracle 19c+)	Up to 30+ PDBs
RAC	No	No (removed 19c+)	Yes
Advanced security	No	Limited	Full (TDE, VPD, Audit)
Flashback	No	Limited	Full point-in-time
Data Guard	No	Basic	Full
Partitioning	No	No	Yes
Production use	No	Small systems only	Enterprise-grade

Key Oracle Architecture Concepts

Container Database (CDB)	The physical Oracle instance — owns memory (SGA), background processes, and redo logs. Acts as the 'building' hosting multiple isolated databases. One CDB runs as one Oracle process but serves many tenants.
Pluggable Database (PDB)	A self-contained, portable database inside a CDB. Looks like a fully independent database to applications. PDBs can be cloned instantly, unplugged, moved between CDBs, and backed up independently — ideal for multi-tenant SaaS.
PL/SQL	Oracle's procedural extension to SQL. Supports stored procedures, packages, triggers, cursors, and exception handling. Business logic runs inside the DB engine — eliminating network round-trips. This is why migrating away from Oracle is expensive.
Oracle SQL Developer	Oracle's free graphical IDE for DBAs and developers. Provides SQL worksheet, PL/SQL debugger, schema browser, data modeling, and PDB management. Equivalent to SSMS (SQL Server) or pgAdmin (PostgreSQL).

Why Oracle dominates enterprise: Oracle's RAC runs one database across multiple physical servers simultaneously. If one server fails, others continue — zero interruption. No other RDBMS matches this for 24/7 critical systems. Banks and telecoms pay millions because downtime costs more in fines and lost revenue.

Part 3 — Enterprise Decision Thinking

Why does a bank choose Oracle over MySQL?	Banks process millions of transactions daily with zero tolerance for data loss. Oracle's RAC, row-level auditing, TDE encryption, and regulatory compliance certifications (SOX, Basel III) are legal requirements. MySQL cannot provide these guarantees natively.
Why does a startup choose PostgreSQL over Oracle?	A startup with limited budget cannot afford Oracle licensing, hardware, or a specialist DBA. PostgreSQL is free, deploys to AWS in minutes, handles JSON and geospatial data natively, and scales with the product. It covers 90% of Oracle's capabilities at nearly zero cost.
Why do companies stay with SQL Server?	Organizations built on Windows, Azure, Active Directory, and Power BI have deep integrations that are expensive to replace. Retraining staff and rewriting BI pipelines (SSIS, SSRS) costs more than the SQL Server license. Rational economics, not inertia.
How does cloud affect database decisions?	Cloud managed services (AWS Aurora, Azure Database for PostgreSQL) provide automated backups, failover, and scaling without a DBA — removing one of Oracle's traditional advantages. Companies choose PostgreSQL/MySQL for new cloud workloads while keeping Oracle only for legacy on-premise systems.
Why pay for commercial DBs when free ones exist?	In regulated industries, software selection is a compliance matter. Auditors require a signed SLA from a named vendor with legal accountability. 'The open-source community will help us' carries no legal weight in a financial audit. Oracle and Microsoft provide 24/7 support, compliance docs, and certified security patches.

Part 4 — Architecture & Scalability Analysis

Architecture Comparison Across Critical Dimensions

Dimension	Oracle	SQL Server	PostgreSQL	MySQL
Scalability	Excellent — RAC active-active multi-node	Very good — AlwaysOn AG	Good — needs Patroni/Citus	Moderate — read replicas only
Security	Industry-leading — VPD, TDE, Audit	Strong — AD, TDE, Always Encrypted	Good — RLS, pgcrypto, SSL	Basic — TLS, user permissions
Licensing	Commercial — \$47K+/processor	Commercial — \$14K+/core	Free — BSD license	Free — GPL community
Enterprise support	24/7 Oracle SLA — legally binding	24/7 Microsoft SLA	Community + EDB (paid)	Community + Oracle/Percona
Cloud readiness	OCI native, AWS, Azure (complex)	Azure native, AWS, GCP	All clouds — best fit	All clouds — widely supported
Backup & recovery	RMAN + Flashback — per PDB	Native backup + AG zero-loss	pg_dump, PITR via WAL	mysqldump, binary log PITR
Stored procedures	PL/SQL — most powerful	T-SQL — very capable	PL/pgSQL — flexible	Basic stored procedures
Integration	Oracle ERP, Siebel, JDE, OCI	Microsoft 365, Azure, Power BI	Open APIs, all frameworks	LAMP, WordPress, CMS

Which DBMS Is Used Where?

Environment	Primary DBMS	Why
Enterprise systems	Oracle, SQL Server	Vendor SLA, compliance certs, long-term support contracts
Startups / MVPs	PostgreSQL, MySQL	Zero licensing cost, fast deployment, cloud-native
Government / public sector	Oracle, SQL Server	Regulatory requirements, certified security, procurement contracts
Cloud-native applications	PostgreSQL, MySQL	Managed services, auto-scaling, no DBA overhead
Core banking systems	Oracle	RAC high availability, advanced auditing, PL/SQL transaction control
Large transactional OLTP	Oracle, SQL Server	Mature query optimizer, proven performance at billion-row scale

Part 5 — Migration & Industry Reflection

Why don't companies use one universal DBMS? All four systems support tables, joins, triggers, procedures, and transactions — but the SQL surface hides completely different internals: concurrency models, locking strategies, optimizer behavior, security architectures, and failure modes. Choosing a DBMS shapes architecture, staffing, compliance, and cost for 10–20 years.

Why Companies Migrate & Key Challenges

Migration Trigger	Risk	Technical Challenge	Business Impact
Cost pressure	HIGH	Schema & data type mapping, PL/SQL rewrites	Oracle at 100 cores = \$2.4M/yr. PostgreSQL migration saves 60–80%
Cloud-first mandate	HIGH	Application SQL rewrites, OCI vs AWS	Oracle licensing on AWS is complex. PostgreSQL Aurora integrates
End of vendor support	MEDIUM	Version upgrade vs full migration decision	Oracle 11g end-of-support forces \$M upgrade or migration project
Vendor lock-in risk	MEDIUM	Identifying proprietary SQL syntax dependencies	Bosch mandates multi-vendor strategy to reduce single-supplier exp
Talent / skills	LOW	Team retraining, documentation	PostgreSQL engineers more available and cost less than Oracle DB

Why Migration Is High-Risk

Risk Area	What Goes Wrong
Schema incompatibilities	Oracle data types (NUMBER, VARCHAR2, DATE with time) don't map directly. Every table must be reviewed. Silent p
Stored procedure rewrite	PL/SQL cannot auto-convert to PL/pgSQL or T-SQL. 500 packages = 12–18 months of engineering work. Each must b
Application dependencies	ROWNUM, CONNECT BY, Oracle MERGE syntax embedded in application code must all be identified and rewritten.
Cost overruns	Enterprise migrations average \$875,000 total cost with \$250,000+ overruns. 80% miss original deadline or budget (ind
Downtime risk	Live production migration requires dual-write strategies and rehearsed cutovers. Even well-planned migrations carry u
Compliance gaps	During transition, audit trail or security controls may be temporarily incomplete — creating a regulatory risk window.

Architecture conclusion: Strong engineers choose databases based on lifespan, regulatory environment, team capability, and total cost of ownership — not personal preference. A 20-year Oracle banking system may be impossible to justify migrating. A new SaaS product should almost always start on PostgreSQL. Database decisions compound over time and shape what is possible — and what is expensive — for the next decade.